FoldPulse

Every Fold. Every Beat.

PHASE 4 · MVP DEVELOPMENT

Development Report

Version	1.0
Date	9 November 2026
Prepared for	Founder & Product Owner
Prepared by	FoldPulse product team (12 agent roles)
Status	Submitted for Founder approval
Recommendation	**APPROVE** (open items P4-1, P4-2)

Confidential — prepared for the FoldPulse Founder. Screenshots show sample data; build results are from CI.

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1 Executive summary & recommendation

PHASE 4 RECOMMENDATION

APPROVE

The MVP is implemented and CI is green. Two open items need your Mac and iPhone Duo at the start of Phase 5.

What was built

The FoldPulse MVP (lean v1, decision D-6) is implemented in Swift 6 with complete strict concurrency:

Feature	Status
F1 Automatic counting while the app is open (hinge)	Implemented. Logic tested in CI; the 15-line iOS 27.1 hookup builds on your Mac (P4-1)
F2 Background counting via the Shortcuts automation (App Intents, background)	Implemented; compile-verified. Behaviour on device = Phase 5 test S-2/S-3
F3 Guided setup + self-test (60 s window, names the failing step)	Implemented and tested
F4 Manual recording with Undo	Implemented; UI-tested end to end
F5 Today dashboard (honest "counted" label, "counting only while open" card, Last row)	Implemented
F6 History (filters, day sections, detail, edit, delete)	Implemented
F7 Insights (day/week/month/year, previous period, busiest day, full-day averages)	Implemented and tested
F8 Milestones (once, ever; optional local notifications)	Implemented and tested
F9 Widgets (Home small/StandBy, Lock Screen circular and rectangular)	From Phase 3, polished (privacy, empty state)
F10 Export CSV/JSON (injection-safe, written on request, removed afterwards)	Implemented and tested
F12 Reset (type RESET)	Implemented and tested
F13 Settings (theme, notifications, privacy)	Implemented
F14 Gap detection (shown, not counted)	Implemented and tested
F15 Coverage ("Counted for you · % of recorded folds")	Implemented and tested
Onboarding, manual-only mode	Implemented

How it was verified

- **89 automated tests, 0 failures** (CI run #14, commit `80e99c3`):
  - **57 core tests** on Linux and macOS;
  - **29 app tests** with SwiftData;
  - **3 UI tests** on the iPhone 17 Pro simulator, including recording a fold end to end and capturing the screenshots in Chapter 2.
- A **privacy guard** in CI fails the build if networking code appears.
- **Two independent reviews** found 4 High and 13 Medium issues, all traced by hand, including:
  - a phantom fold on returning to the app;
  - a lost fold after an "unknown" reading;
  - de-duplication and gap errors caused by the 7-day window;

  - double counting from lagging Shortcuts;
  - export files left behind;
  - Undo timing out for Voice Control users.

  All High and Medium findings are fixed with regression tests, except the two items that need your Mac (P4-1, P4-2).

- CI also caught real problems: one Swift 6 compile error in a test, and a UI bug where the milestone celebration covered another tab.

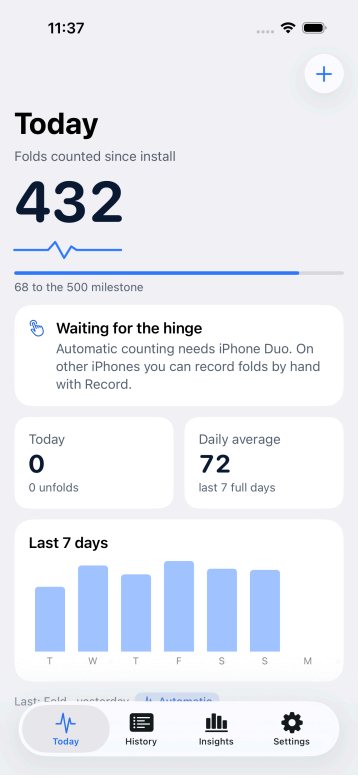
Recommendation

Approve Phase 4. Open items, recorded and not hidden:

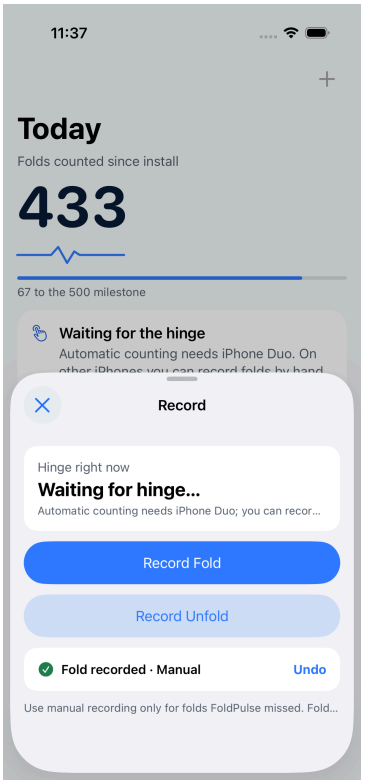
1. **P4-1:** the hinge hookup must be compiled with Xcode 27.1 and run on your iPhone Duo. That's the first Phase 5 step.
2. **P4-2:** the inner-display `ArrangementView` split and the half-folded tab bar (I-24) need the iPhone Duo simulator or device.
3. **R-28:** background counting through Shortcuts is still unverified on a device (Phase 5 gate).
4. **R-29:** the Apple Developer Program is likely needed for the App Group on your device.

2 The app, running

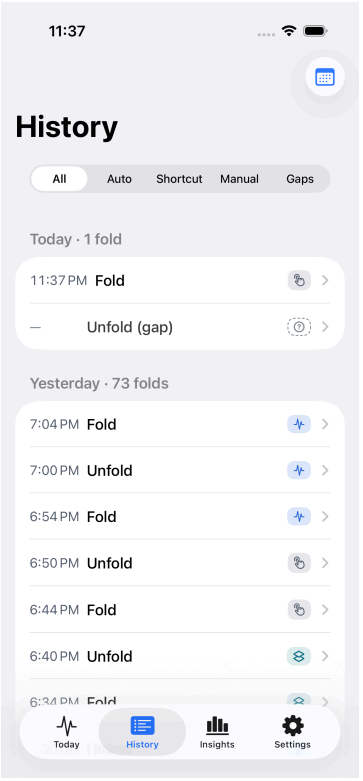
Real screenshots from the **running app**, captured by the UI test `ScreenshotTests` in CI. Simulator iPhone 17 Pro, iOS 26.5 SDK, debug build with the sample week. They are not mockups. The "Waiting for the hinge" card is expected: the simulator has no hinge. These screenshots revealed one layout bug, truncated text in the Record sheet at its medium height. It was fixed right after capture: the sheet now scrolls and the text wraps.



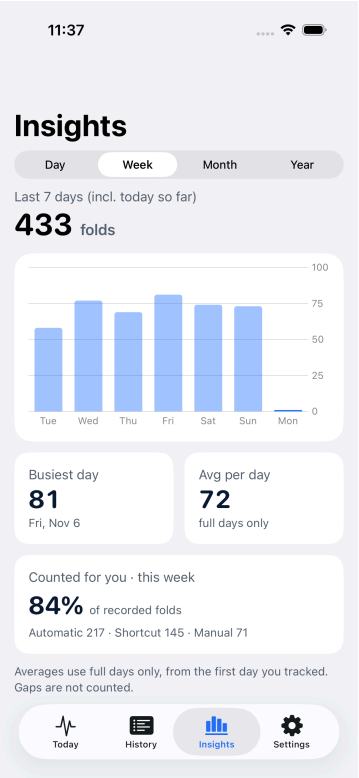
Today (sample week)



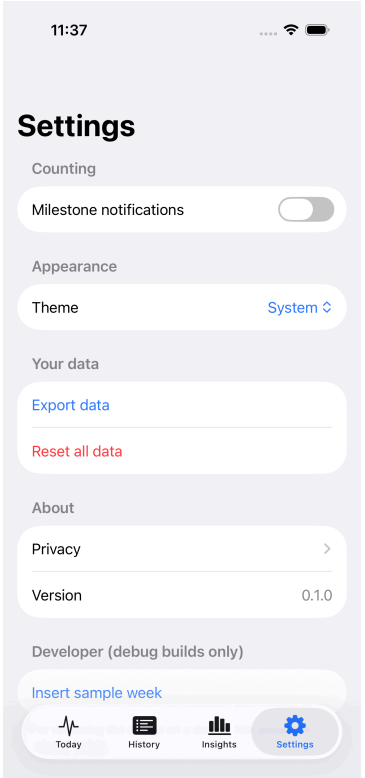
Record sheet with Undo



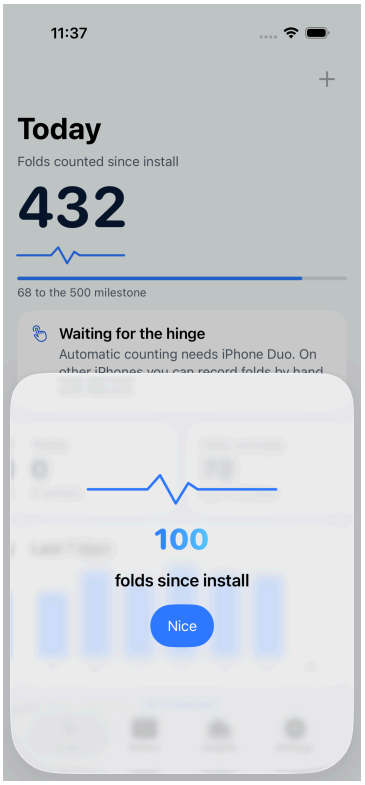
History



Insights · week



Settings



Milestone celebration (once)

3 Development log

Each entry follows the brief: plan → responsible agent → implementation → build → tests → review → fixes → docs → result. Build results come from CI (GitHub Actions); run numbers link to https://github.com/odthientho/FoldPulse/actions.

Working method

- Features were built in **three batches**, each verified by CI before the next was trusted:
  - **A** — counting engine (pure Swift);
  - **B** — recording;
  - **C** — screens. Batches B and C were pushed together but committed separately (`21efabf` , `6b0dddb`).
- **Hinge code** (`HingeTrackingModifier`) is compiled only with the iOS 27.1 SDK, behind `HINGE_API` (DECISIONS T-10). Its logic lives in `HingeTracker` and `HingeEdgeDetector` , which CI does test. The SwiftUI hookup uses the exact API form that compiled and ran in FoldProbe on the Founder's iPhone Duo.
- **Independent reviews** ran on the complete Phase 4 code (see "Reviews and fixes").

Batch A — counting engine (commit `870359d` , CI run #7: green on Linux and macOS)

Feature	Plan	Responsible	Implementation	Tests
F1 rules: hinge edges	Turn raw hinge readings into folds/unfolds per feasibility §6	Agent 7 (rules), Agent 6 (code)	`HingeEdgeDetector` : 250 ms debounce, first-entered timestamp, baseline-on-activation, flush, nil resets baseline	7 tests: baseline, debounce, bounce, many angle callbacks, late tick, flush, resume-without-false-event
De-duplication and gaps	One physical fold → one event, regardless of source or order; gaps when alternation breaks	Agent 7, Agent 6	`Reconciler.plan` : ±10 s either order, precedence hinge > shortcut > manual, one-to-one per source, same-source 1 s, gap interval bucketed on the later day, idempotent	11 tests: both orders, window edge, opposite event in between, manual-after-automatic, one-to-one, idempotence, delete restores duplicate, gap removed when filled, gap day
F7 Insights data	D/W/M/Y with honest averages	Agent 7	`Statistics.summary`	4 tests: hourly, week (full-day average, busiest day), month/year with previous period, empty
F10 Export	Safe CSV + full-fidelity JSON	Agent 8 (rules), Agent 6	`Exporter` : formula-injection escaping, tombstones omitted from CSV, ISO-8601 milliseconds	3 tests: injection cases, ordering/tombstones, JSON round trip

Batch B — recording (commit `21efabf`)

Feature	Plan	Responsible	Implementation	Tests
Shared services	One writer for the UI and background intents	Agent 3, Agent 6	`AppServices` (store, snapshot, settings), `AppSettings` (App Group defaults)	Covered by recording tests
Recording + reconciliation	Every write reconciles a 7-day window	Agent 6	`EventStore.record/edit/delete/reconcile`	`RecordingTests` : hinge+shortcut count once; missed unfold → gap, not counted; Undo; edit moves the day; milestone once
F1 Automatic counting	Count while on screen	Agent 6	`HingeTracker` (ordered delivery, debounce timer, flush) + `HingeTrackingModifier` (HINGE_API)	`HingeTrackerTests` (3)
F2 Background counting	Shortcuts automation → App Intent in background	Agent 6, Agent 10	`RecordFoldIntent` / `RecordUnfoldIntent` (`supportedModes = .background`), App Shortcuts	Compile-verified in CI; behaviour on device = Phase 5 (S-2/S-3)

Feature	Plan	Responsible	Implementation	Tests
F4 Manual + Undo	Record by hand, undo for 5 s (persistent with VoiceOver/Switch Control)	Agent 5	`RecordSheet`	UI test (batch C)
F8 Milestones	Celebrate once, ever; optional local notification	Agent 4, Agent 6	`AppSettings.claimMilestones`, `MilestoneSheet`, `MilestoneNotifier`	`AppSettingsTests`, `RecordingTests`

Batch C — screens (commit `6b0dddb`)

Feature	Implementation	Tests
F6 History	Source filter (segmented Picker), range Menu, day sections, swipe delete (confirm for automatic), detail, edit sheet	Store queries tested; UI test checks the recorded fold appears
F7 Insights	D/W/M/Y chart, previous period, busiest day, full-day averages, coverage	Core summary tests
F15 Coverage	Detail per source, gaps, background status	Core coverage tests
F3 Setup + self-test	Plain steps, iOS 27.2 check, Open Shortcuts, self-test (both automations after start, within 10 min), troubleshooting	Logic in `AppModel.checkSelfTest` (on device: Phase 5)
F12/F13 Settings	Theme, milestone notifications (permission on enable), export CSV/JSON with privacy note, two-step reset (type RESET), privacy statement	Export covered by core tests
Onboarding / manual-only	3 pages on first launch; "Waiting for the hinge" notice until a reading is seen	Skipped in UI tests (`-ui-testing`)
UI test	Record → Record Fold → Today shows "1 fold since install" → History lists it	`LaunchTests` (2)

CI history (Phase 4)

Run	Commit	Result	Notes
#7	`870359d` (batch A)	**Green**	Core tests pass on Linux and macOS
#8	`6b0dddb` (batches B + C)	Failed	One compile error in a test: `XCTUnwrap(await …)` isn't allowed (async call in a non-async autoclosure). All app sources compiled
#9	`1436d83`	**Green**	Batches B + C build and pass all tests (pre-review)
#11, #12	`53f8922`, `e2abeac`	Cancelled	Superseded while still queued (GitHub keeps one waiting run per branch); their code is included in #13
#13	`7249679`	Failed (UI)	All 86 unit/core tests and both functional UI tests pass. The new `ScreenshotTests` found a **real UI bug**: the milestone sheet presented over the Settings tab and covered the tab bar
#14	`80e99c3`	**Green**	Milestone sheet only while Today is visible. **57 core + 29 app + 3 UI = 89 tests, 0 failures**

Screenshots and the last UI fix

Run #16 (`c8d0aa8`, green) published real screenshots of the running app to the `ci/screenshots` branch (see the Phase 4 report, Chapter 2). They showed truncated text in the Record sheet at its medium height. The sheet now scrolls and its text wraps; this is verified by the run after `c8d0aa8`.

Reviews and fixes

Two independent reviews ran on the complete Phase 4 code (see PHASE4_REVIEWS.md):

- **Engineer + Data Analyst:** 4 High, 3 Medium, 3 Low.
- **Privacy + Compliance + QA + Independent:** no Critical, conditional approval.

The fixes are in commit `53f8922`, with regression tests:

- **Rule change (Data Analyst):** de-duplication now uses hinge events as the backbone, and lower-precedence events pair one-to-one with the nearest kept event within ±10 s.

- **Hinge baseline:** reset on activation, and "unknown" never changes it.
- **Reconciliation:** slice protection.
- **Background-task protection** for saves.
- **Store-unavailable errors** for Shortcuts.
- **Privacy:** export/temp-file hygiene and notes cleared on delete.
- **Accessibility:** Undo persistence, Dynamic Type, Reduce Motion.
- **Missing Today states.**
- **Self-test:** fail state.

Open items carried to Phase 5

- **P4-1:** compile and run the hinge hookup with Xcode 27.1 on an iPhone Duo.
- **P4-2:** inner-display `ArrangementView` split and the half-folded tab bar (I-24).
- **P4-3:** App Store wording, "Requires iPhone Duo for automatic counting" (Phase 6).
- **P4-4:** v1.1 backlog (DST-normalised heat strip, custom History range).

4 Agent reviews & resolutions

How the reviews were run

Two independent reviewer subagents read all the Phase 4 code, tests and specifications. They could not edit files or compile; CI compiled separately. Each finding came with a hand-traced failure scenario.

Reviewer	Roles	Verdict
A	Agent 6 — Senior iOS Software Engineer; Agent 7 — Data Analyst	7 counting/engineering issues (4 High)
B	Agent 8 — Privacy; Agent 10 — Compliance & Accessibility; Agent 9 — QA; Agent 12 — Independent	No Critical; **conditional approval**

Every High and Medium finding was fixed in commit `53f8922`, with regression tests, unless listed under "Not fixed in Phase 4".

Engineer (Agent 6)

#	Finding	Sev.	Resolution
E1	The hinge baseline survived a trip to the background. Opening FoldPulse hours later on the outer display created a **phantom fold**	High	Baseline reset when the scene becomes active or background; test `testReturningToTheAppNeverCreatesAPhantomFold`
E2	An "unknown" status reset the baseline, so a fold could be lost	High	Unknown updates the label only; test `testUnknownNeverChangesTheBaseline`
E3	Every window fed one shared detector (iPhone Duo supports several)	Medium	Only the foreground-active scene counts
E4	A fold flushed as the phone closes could be lost when the app is suspended	Medium	Each save runs inside a background task
E5	A Shortcut "succeeded" while the store was unavailable, so the fold was lost and the self-test could pass falsely	Medium	`record` throws `StoreUnavailableError`; tested
E6	Swipe-deleting a gap brought it back	Low	Gaps can't be swipe-deleted

Data Analyst (Agent 7)

#	Finding	Sev.	Resolution
D1	The 7-day reconcile window cut through duplicate pairs and gaps: duplicates resurfaced and long gaps were deleted about a week later	High	Slice protection (`protectBefore`); test `testWindowProtectionKeepsPairsAndGapsThatStartBeforeTheSlice`
D2	Lagging Shortcuts after a quick close-and-reopen double counted (2 folds instead of 1)	Medium	**Rule change:** hinge events are the backbone; Shortcut, then manual, events pair one-to-one with the nearest higher-precedence kept event within ±10 s, even across an opposite event. Shortcut events never pair with each other. Tests for both cases
D3	A double tap next to a hinge event counted twice	Medium	Same-source repeats grouped first; tested
D4	Replacement chains stretched the 10 s window	Low	Removed by the new pairing (no chains)
D5	The Day chart puts both copies of the repeated DST hour in one bar	Low	**Accepted** for the single-day chart; the time-of-day heat strip (v1.1) will normalise

Privacy & Security (Agent 8)

Finding	Sev.	Resolution
Export files stayed in tmp; the file was written as soon as the sheet opened	Medium	Written only on request; removed on close, launch and reset; tested
Deleted events kept their note and were exported	Medium	Deleting clears the note; export omits deleted events; tested

Finding	Sev.	Resolution
Lock Screen gauge not fully private	Low	`.privacySensitive()` on the whole gauge
Reset left delivered notifications; the CI privacy guard missed some networking APIs	Low	Both fixed
P-1 (no networking), P-2 (manifests), local-only notifications, CSV injection	—	Confirmed met

Compliance & Accessibility (Agent 10)

Finding	Sev.	Resolution
Undo disappeared after 5 s for Voice Control and Full Keyboard Access users (can't be detected)	Medium	Undo now stays for everyone until the sheet closes or the next record
Dynamic Type truncation (fixed widths and sizes)	Medium	`@ScaledMetric` and text styles throughout
The app is paid, but only manual counting works on non-Duo iPhones	Medium (App Review)	**Phase 6:** the App Store text must state "Requires iPhone Duo for automatic counting" (guidelines 2.3, 2.4.1)
Reduce Motion ignored in two places; haptic also fired when the toast hid; missing success haptic; chart bars spoken as ISO dates	Low	All fixed

QA (Agent 9)

Finding	Sev.	Resolution
The hinge mapping code is never compiled in CI (needs the iOS 27.1 SDK)	High	The mapping is now a pure `HingeReading` path tested in CI. The remaining 15-line SwiftUI hookup **must be compiled on the Founder's Mac (Xcode 27.1) before Phase 5**: open item **P4-1**
Background-save risk; intent fallback	Medium	Fixed (E4, E5)
History silently capped at 2,000	Medium	Limit raised to 10,000, with an on-screen notice when reached. The 50,000-event scroll test is a Phase 5 performance test
Untested self-test, export, reset, intents	Medium	Self-test, export and reset now tested. The intent bodies call the tested `AppServices.record` ; their run in Shortcuts is the Phase 5 device test S-2/S-3

Independent Final Reviewer (Agent 12)

Finding	Sev.	Resolution
Today lacked the "Folds counted since install" label, the "Counting only while FoldPulse is open" card and the "Last:" row	Medium	Implemented
Self-test had no fail state; window was 600 s, not 60 s	Medium	60 s window; the failure names the missing automation; tested
"Working" showed even after days without a run	Medium	"No automation run in 24 h"
Inner-display `ArrangementView` split and fold avoidance not implemented	Medium	**Not fixed in Phase 4**: needs the iOS 27.1 SDK and the iPhone Duo simulator, which only the Founder's Mac has. Combined with I-24 as open item **P4-2** for the Phase 5 device session. Until then the inner display uses standard full-width layouts, which the system already adapts to the fold
Widget "—" vs spec "0" at midnight	Low	Spec corrected to "—" (honest until refreshed)
Widget empty state; update notice should replace the steps; hide coverage/setup on manual-only devices; README	Low	All fixed
Custom History date range; Gaps filter always shown	Low	v1.1 backlog
Scope	—	No scope creep; debug tools excluded from release builds

Not fixed in Phase 4 (open items)

ID	Item	Plan
P4-1	Compile and run the hinge hookup with Xcode 27.1 on an iPhone Duo	Founder's Mac, start of Phase 5 (one-line setup script)
P4-2	Inner-display `ArrangementView` split; inner tab bar when half-folded (I-24)	Phase 5, on the iPhone Duo simulator / device
P4-3	App Store text: "Requires iPhone Duo for automatic counting"	Phase 6
P4-4	DST-normalised hourly heat strip, custom History range	v1.1

5 Architecture (updated)

1. Principles

1. **The event log is the source of truth.** Every number is computed from it, and nothing derived is stored as authoritative (T-1).
2. **Pure domain logic lives in its own package** (`FoldPulseCore`). It has no UIKit, SwiftUI or SwiftData, so it is deterministic and testable on Linux and macOS.
3. **One database writer.** All persistence goes through the `EventStore` actor (T-2).
4. **Local only.** There is no networking code; CloudKit is explicitly off (T-3).
5. **Widgets never open the database.** They read a small JSON snapshot (WIDGET_DESIGN §2).
6. **System components first.** `TabView` , `List` , `Chart` and `Gauge` let the system handle iPhone Duo layouts, Liquid Glass and accessibility.
7. **Swift 6 with complete strict concurrency** in every target.

2. Module structure

```
FoldPulse (repo)
├── project.yml                  XcodeGen spec → FoldPulse.xcodeproj (generated, not committed)
├── Packages/FoldPulseCore/      Swift package — domain logic (pure Swift)
│   ├── Sources/FoldPulseCore/   FoldEvent, EventKinds, LocalDay, Statistics, Milestones, WidgetSnapshot
│   └── Tests/                   XCTest — runs on Linux and macOS
├── Shared/                      compiled into BOTH app and widget
│   ├── Storage/                 AppGroup (paths), SnapshotFile (read/write + file protection)
│   └── DesignSystem/            Tokens (colours, spacing, radii), PulseLine
├── FoldPulse/                   app target
│   ├── App/                     FoldPulseApp (entry), RootView (tabs), AppModel (state)
│   ├── Core/Persistence/        SchemaV1 (@Model), mapping, PersistenceController, EventStore (@ModelActor)
│   ├── Core/Services/           SnapshotService (widget snapshot + reload policy)
│   ├── DesignSystem/            Typography, components (HeroCounter, StatTile, SourceBadge, …)
│   ├── Features/                Today · History · Insights · Settings
│   └── Resources/               Assets (icon, accent), PrivacyInfo.xcprivacy, entitlements
├── FoldPulseWidgets/            widget extension (TodayWidget: small, circular, rectangular)
├── FoldPulseTests/              app unit tests (in-memory SwiftData)
├── FoldPulseUITests/            UI smoke tests
├── scripts/setup_mac.sh         one-line developer setup
└── .github/workflows/ci.yml     CI: Linux core tests + macOS build & tests
```

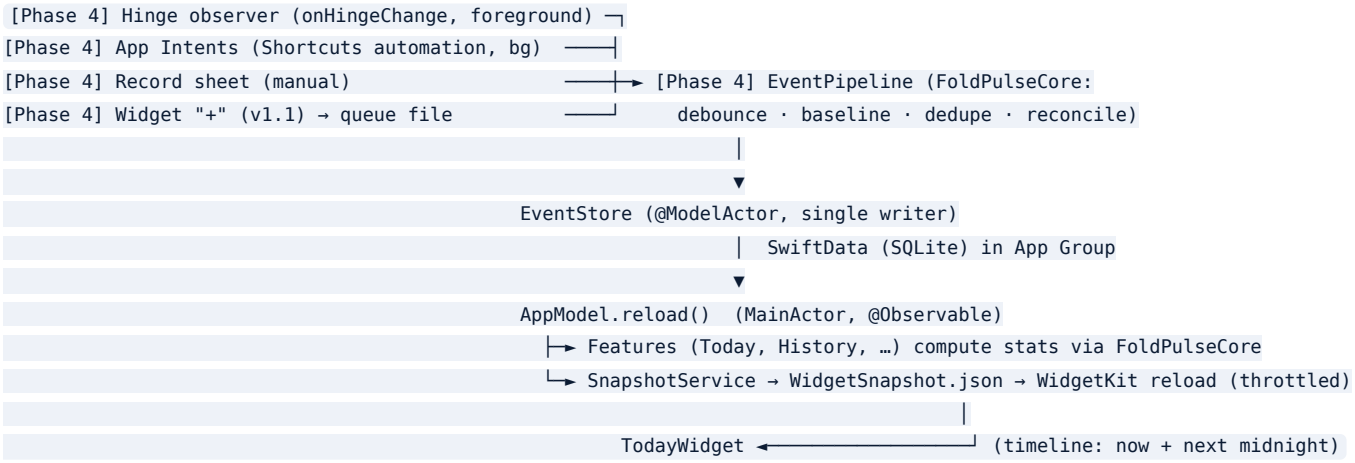
Dependency rule: **Features → AppModel → EventStore / SnapshotService → FoldPulseCore**. Shared depends only on FoldPulseCore and Apple frameworks. The widget depends on Shared + FoldPulseCore, never on app code.

3. Targets and capabilities

Target	Type	Bundle ID	Capabilities
FoldPulse	iOS app (iPhone)	`io.github.odthientho.foldpulse`	App Group `group.io.github.odthientho.foldpulse`
FoldPulseWidgets	WidgetKit extension	`…foldpulse.widgets`	Same App Group
FoldPulseTests	Unit tests	`…foldpulse.tests`	—
FoldPulseUITests	UI tests	`…foldpulse.uitests`	—

Deployment target **iOS 27.1** (D-3). Mac ("Designed for iPhone") and visionOS availability are **off** (R-06). Category: Utilities (R-27).

4. Data flow



The diagram for the PDF is Figure 2 of the Phase 3 report.

5. Concurrency model

Component	Isolation	Notes
SwiftUI views, `AppModel`, `SnapshotService`	`@MainActor`	`AppModel` is `@Observable` and injected with `.environment`
`EventStore`	`@ModelActor` (its own serial executor), **created on a detached task** so its context isn't bound to the main thread	Exchanges only `Sendable` values (`FoldEvent`, `LoadedState`); computes `DashboardStats` itself, so statistics never run on the main thread
`FoldPulseCore`	Non-isolated, pure value types	All types `Sendable`
Widget provider	WidgetKit-managed	Reads a file; no shared mutable state

6. Persistence

See **DATABASE_DESIGN.md**. In summary:

- SwiftData `SchemaV1.StoredEvent` with a `SchemaMigrationPlan` from day one.
- The store file lives in the App Group container.
- Rows are tombstoned instead of deleted.
- File protection is `completeUntilFirstUserAuthentication`, **set explicitly** on the store directory, so background writes work while the phone is locked.
- Release builds never fall back to a private location if the App Group is missing; they show a clear error instead.

7. Widget pipeline

- `SnapshotService.publish` writes the precomputed `WidgetSnapshot` atomically and calls `WidgetCenter.reloadTimelines(ofKind:)`. That reload is immediate when the app is active. From the background it is coalesced to at most one per 5 minutes; the last-reload time is kept in the **App Group defaults**, so the throttle holds across short-lived intent processes.
- The `TodayWidget` timeline has up to three entries: now; the moment the data turns **stale** (shows "as of HH:mm"); and the **next local midnight** ("today" shows "—" until refreshed). The widget also re-reads the snapshot **every 30 minutes**, so a reload the app throttled is never lost.

8. Testing strategy

Layer	Framework	Where it runs	Phase 3 coverage
Domain (`FoldPulseCore`)	XCTest via `swift test`	**CI Linux + CI macOS**	LocalDay (DST, travel, leap year, ranges), totals (gaps, tombstones, duplicates), averages (full days, zero days, window cap), coverage, busiest day, milestones (fire once), snapshot (encoding, schema guard, midnight, staleness)
Persistence & services	XCTest, in-memory SwiftData	CI macOS (simulator)	Field-preserving round trip, ordering, update, tombstone, not-found, delete-all, snapshot write and reload throttling
UI	XCUITest	CI macOS (simulator)	Launch into Today (empty state), all four tabs reachable
On device	Manual	Founder's Mac + iPhone Duo	Sample week via the debug-only Settings action

9. Build and run

- **Mac:** `curl -fsSL https://raw.githubusercontent.com/odthientho/FoldPulse/main/scripts/setup_mac.sh | bash`, then choose a Team for both targets and press Run.
- **CI:** every push to `main` runs `.github/workflows/ci.yml`.
  - On Linux: core tests in the `swift:6.1` container.
  - On macOS: the newest Xcode; core tests, then XcodeGen, then `xcodebuild test` on a simulator.
  - If the runner's iOS SDK is older than 27.1, CI lowers the deployment target in `project.yml` **for CI only** and says so in the job summary. GitHub's runners currently have the **iOS 26.5 SDK**. Phase 3 code uses no iOS 27-only API, so build and tests are meaningful there. From Phase 4, hinge code needs the 27.1 SDK; it will be compiled conditionally or verified on the Founder's Mac.
  - A **privacy guard** fails the build if networking symbols appear in shipping Swift code (PRIVACY_REVIEW P-1).

10. Phase 4 extension points

Feature	Where it goes
Event pipeline (debounce, baseline, symmetric dedupe ±10 s, gap reconciliation)	`FoldPulseCore/Pipeline` — pure, unit-tested on Linux
Hinge observer	`FoldPulse/Core/Services/HingeObserver` (one per process, foreground scene only)
App Intents (Record Fold/Unfold, `.background`)	App target, writing through `EventStore`
Manual record / Undo, History edit/delete, Insights, Coverage, Setup + self-test, Export, Reset, Milestones, Onboarding	`Features/*`

10a. Phase 4 implementation (as built)

Concern	Where	Notes
Counting rules	`FoldPulseCore/Pipeline/HingeEdgeDetector`, `Reconciler`	Pure, tested on Linux + macOS. Reconciler: hinge backbone, one-to-one pairing of lower-precedence events within ±10 s, same-source groups, gaps, slice protection
Automatic counting	`HingeTracker` (+ `HingeTrackingModifier` under `HINGE_API`)	Active scene only; baseline reset on activation; unknown ignored; saves in a background task
Background counting	`Intents/RecordIntents.swift` → `ShortcutRecorder` → `AppServices.record`	`supportedModes = .background`; fails visibly if the store is unavailable
Shared services	`AppServices` (store, snapshots, settings)	Used by the UI and by intents; one writer (`EventStore`)
Reconciliation	`EventStore.record/edit/delete → reconcile(around:)`	7-day slice per write, protected boundary
Settings and state	`AppSettings` (App Group defaults)	Milestones claimed once, self-test, last automation runs

Concern	Where	Notes
Screens	`Features/` Today, Record, History (+ detail/edit), Insights, Coverage, Setup, Settings (+ export/reset/privacy), Onboarding, Milestone	Statistics computed in `EventStore`, off the main thread

11. Known limitations and open items

ID	Item	Plan
I-24 / P4-2	Inner-display tab bar position when partially folded; `ArrangementView` split for the inner display	Check and implement in Xcode 27.1 Device Hub on the Founder's Mac (Phase 5)
P4-1	The `HINGE_API` hookup is compiled only with the iOS 27.1 SDK	Founder's Mac build at the start of Phase 5
R-29	**App Groups may need a paid Apple Developer Program membership** for device installs (free personal teams have restricted capabilities)	If signing fails on the Founder's Mac, either join the Program ($99/year, also required for App Store release) or temporarily remove the App Group (widgets then show the empty state)
R-01	No iPhone Duo in CI	On-device checks stay with the Founder

6 Decisions requested & next steps

Decisions requested

#	Decision	Team recommendation
A-4	**Approve Phase 4** (MVP implemented; CI green; reviews' High and Medium items fixed)	Approve
P4-1	At the start of Phase 5: build on your Mac with Xcode 27.1 and run on your iPhone Duo (one-line setup script), so the hinge code is compiled and exercised for real	Required
R-29	Apple Developer Program ($99/year): likely needed for the App Group (widgets) on your device, and required for release	Join before Phase 5 device testing
R-28	Background counting on iOS 27.2 (spike S-2/S-3) is still unverified	Phase 5 gate

Phase 5 — Quality assurance & validation (after approval)

1. **On your iPhone Duo (with you):**
   - P4-1 build;
   - the spike protocol S-1…S-7, re-run with the real app: accuracy over at least 300 cycles, the automation while locked, rotation, battery;
   - widgets on the Lock Screen and in StandBy;
   - P4-2 inner layout and I-24.
2. **Functional, data, widget, privacy and performance testing:**
   - time-zone and DST runs;
   - persistence across restarts;
   - a 50,000-event History scroll;
   - export with corrupt-data checks;
   - proxy capture showing zero network requests.
3. **Deliverables:** TEST_PLAN.md, TEST_RESULTS.md, BUGS.md, the privacy and performance reviews, the release-blocking list, and `phase-05-testing.pdf`.